Spatial modeling improves the calibration of leaf wax hydrogen isotopes to precipitation Supplementary Material

S1.1 Ordinary Least Squares Regression

The relationship between δ^2H_{wax} and δ^2H_{precip} can be described by an ordinary least squares (OLS) regression (Sachse et al., 2012). This approach models δ^2H_{wax} as a linear function of δ^2H_{precip}, with Gaussian noise accounting for unexplained variation around the regression line. It can be written as:

$$\delta^2 H_{wax,i} = \beta_0 + \beta_{\mathrm{OIPC}}\, \delta^2 H_{precip,i} + \epsilon_i$$

(Supplementary Equation 1)

where ($\delta^2H_{precip,i}$ $\delta^2H_{wax,i}$) are data pairs related by the intercept (β_0) and slope β_{OIPC}, and ϵ_i is an error term, typically assumed to be follow a normal distribution:

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

(Supplementary Equation 2)

Equation 1 is a full generative model for δ^2H_{wax} that assumes that all residual variation is unstructured. That is, $\epsilon_i$implicitly absorbs both measurement error and any spatial or ecological effects not included in the linear predictor.

This approach offers several advantages for proxy-based reconstructions (McClelland et al., 2021): it is simple to implement, provides an intuitive means of relating proxy measurements to environmental variables, and enables straightforward estimation of prediction uncertainty. It also helps expose common challenges in proxy calibration, such as the dominance of unexplained variance. However, OLS regression relies on strict assumptions about error structure (e.g., homoscedasticity, uncorrelated residuals). OLS also assumes predictors are measured without error. When this assumption is violated, slope estimates are biased towards zero, a phenomenon known as regression dilution. Consequently, analyses that involve nonlinear responses, interacting covariates, or uncertainty in the predictors require modeling tools that are more flexible than ordinary least squares.

Despite its simplicity, an ordinary least squares (OLS) approach provides a solid baseline for quantifying the δ^2H_{precip}-δ^2H_{wax} relationship. Figure 1 in the main text shows an OLS regression of the δ^2H_{wax} of 818 sedimentary n-C_{29} alkanes against the locally interpolated δ^2H_{precip}. This relationship shows a linear relationship ($p = 2.2 \times 10^{-16}$) with $\beta_0 = -126.9 \pm 1.6$ and $\beta_{OIPC} = 0.802 \pm 0.02$, and an $R^2 = 0.653$, demonstrating that δ^2H_{precip} strongly predicts δ^2H_{wax}.

A critical feature of any calibration is the uncertainty in the predicted relationship. The prediction interval is the appropriate measure for this, as it incorporates both parameter uncertainty and the residual variance (Kraucunas, 2006; McClelland et al., 2021). In an OLS regression the prediction interval incorporates all unmodeled influences on the calibration. Bayesian approaches yield posterior predictive intervals that serve the same purpose, combining model and residual uncertainty. Therefore, whether using frequentist or Bayesian methods, the prediction interval is the most appropriate and informative expression of uncertainty in paleoclimate reconstructions. As emphasized by Kraucunas (2006), the bounds of the prediction interval only capture statistical uncertainty in the calibration dataset, and do not account for additional sources of uncertainty such as non-stationarity. For the regression of δ^2H_{wax} on δ^2H_{precip}, the 95% prediction interval for δ^2H_{wax} exhibits substantial unexplained variability, even near the center of the calibration range. At the mean value of δ^2H_{precip}, the half-width of the prediction interval is 41.6‰ (i.e. $\delta^2 H_{precip,i}$ predicts a value of $\delta^2 H_{wax,i}$ ± 41.6‰), and the full width is double that at 83.1‰.

For applications to paleohydrology, this noisy relationship must be inverted. That is, we measure $\delta^2\mathrm{H}_{\mathrm{wax,0}}$ to make an estimate $\delta^2\widehat{\mathrm{H}_{\mathrm{precip}}}$. Formally, this is done via the inverted equation:

$$\delta^2\widehat{\mathrm{H}_{\mathrm{precip,0}}} = \frac{\delta^2\mathrm{H}_{\mathrm{wax,0}} - \widehat{\beta_0}}{\widehat{\beta_{OIPC}}}$$

(Supplementary Equation 3)

This estimate of δ^2H_{precip} from δ^2H_{wax} carries an even larger uncertainty than the forward calibration. With frequentist methods, an exact solution for the range of the inverse prediction interval (IPI) requires solving a quadratic inequality derived from Fieller's theorem (Fieller, 1954), which fully accounts for uncertainty in the regression slope β_{OIPC}. Although the full Fieller calculation is seldom carried out, an approximation to the IPI can be obtained by assuming no uncertainty in β (McClelland et al., 2021):

$$\delta^2\mathrm{H}_{\mathrm{precip,0}} = \delta^2\widehat{\mathrm{H}_{\mathrm{precip,0}}} \pm \frac{\hat{\sigma}\, t_{\alpha,n-2}}{\left|\widehat{\beta_{OIPC}}\right|}$$

(Supplementary Equation 4)

where $t_{\alpha,n-2}$ is the dimensionless critical value of the Student's *t*-distribution, which depends on *n*, the number of observations in the calibration, and α which determines the confidence level. This equation shows that the uncertainly rises dramatically as the slope $\left|\widehat{\beta_{OIPC}}\right| \rightarrow 0$.

For the global regression shown in Figure 1 of the main text, a δ^2H_{wax} = -180‰ vs. VSMOW would predict a reconstructed δ^2H_{precip} = -64.4‰. The 95% uncertainty range of this reconstructed value calculated by the Fieller exact solution would be [-117.1‰, -12.2‰]; the McClelland approximation would slightly underestimate the uncertainty bounds at [-116.3‰, -12.4‰]. But both methods yield an uncertainty with a full width of more than 100‰ at δ^2H_{wax} = -180‰, and the IPI increases towards the edges of the calibration. This uncertainty range is greater than the variation δ^2H_{wax} signal in many records used for paleohydrological reconstructions (Tierney et al., 2008; Konecky et al., 2016; Daniels et al., 2021; Martins et al., 2022; Zhao et al., 2024).

For this reason, global OLS regressions are unsuitable for reconstructing δ^2H_{precip}. If this method is used, smaller variations in δ^2H_{wax} can be assumed to reflect real ecological or climatic changes, but cannot be assumed to reflect changes in δ^2H_{precip} (McFarlin et al., 2019). In most records, the inference that δ^2H_{wax} directly reflects δ^2H_{precip} is generally not warranted given the magnitude of uncertainty in δ^2H_{precip} reconstructions using global OLS calibrations. By extension, the inference of wetter or drier conditions due to changes in δ^2H_{wax} is also problematic, since detecting a discernable change δ^2H_{precip} requires understanding both the prediction uncertainty and temporal autocorrelation in the δ^2H_{wax} signal, as autocorrelation affects the statistical significance of observed differences between time points. Confidently inferring δ^2H_{precip} or wet/dry cycles requires either additional constraints or alternative approaches (Section 1.3), unless changes are larger in magnitude (McFarlin et al., 2019). Additional constraints could include estimates of the fraction of C_4 vegetation (Collins et al., 2013; Wang et al., 2013; Utida et al., 2019) or of plant functional type (Feakins, 2013; Chang et al., 2021; Daniels et al., 2021; Konecky et al., 2019a).

S2.1 Data Compilation and Processing

S2.1.1 Data Preparation

A global database of 818 published calibration measurements of the n-C_{29} alkane δ^2H_{wax} was constructed by aggregating recent compilations from soils and sediments (Liu and An, 2019; McFarlin et al., 2019; Ladd et al., 2021; Wang et al., 2023; Gaviria-Lugo et al., 2023). We restricted our analysis to the n-C_{29} alkane because this is a widely applied proxy and allows comparability of our model with prior calibrations. However, our method could be applied to other compounds as well.

Environmental predictor variables were extracted from global gridded datasets as described in the main text (Section 2.1.2). All raster processing was performed using R (**terra** package version 1.7) and Python (**earthaccess** for data downloads). The preprocessing workflow consisted of the following steps:

C_4 vegetation: C_4 fractional cover data were extracted from the NUS C_4 distribution dataset NetCDF (Luo et al., 2024). The 19 annual layers (2001-2019) were temporally averaged to create a mean C_4fraction raster. The NetCDF file was transposed and georeferenced to WGS84 to ensure correct spatial orientation.

MODIS land cover: MODIS MCD12Q1 Collection 6.1 land cover data were downloaded for 2001-2019 using the NASA earthaccess API. Annual global mosaics were created from HDF tiles by extracting the LC_Type1 band and mosaicking using GDAL. The sinusoidal projection mosaics were reprojected to WGS84 using nearest-neighbor resampling to preserve categorical class values. A modal land cover classification was computed across the 19-year period, then aggregated to match OIPC resolution (~9 km) while calculating fractional cover percentages for each IGBP class. Finally, IGBP classes were grouped into three plant functional types: trees (classes 1-5: all forest types), shrubs (classes 6-7: shrublands), and grasses (classes 8-10: woody savannas, savannas, and grasslands).

TerraClimate variables: Monthly TerraClimate NetCDF files (Abatzoglou et al., 2018) for precipitation, soil moisture, maximum temperature, and vapor pressure deficit were downloaded for 2001-2019. For each variable and year, monthly values were aggregated to annual values (summed for precipitation; averaged for other variables), then the 19 annual values were averaged to create 2001-2019 long-term means.

All raster datasets were processed in WGS84 (EPSG:4326) coordinate reference system. OIPC δ^2H_{precip} and elevation from GMTED2010 (via EarthEnv-DEM90) were used directly as provided by the data sources. For each sediment sample location, environmental predictors were extracted using a multi-scale spatial averaging approach (Section S2.1.5) with buffers up to 5° radius (approximately 555 km at the equator) rather than point extraction. This integration accounts for the spatial footprint of sediment archives and reduces sensitivity to minor geolocation errors. For C_4 vegetation, pixels with missing data were filled using ecological rules: locations with |latitude| ≥ 50° or elevation ≥ 1500 m were assigned C_4 fraction = 0, consistent with known C_4 plant biogeography (Still et al., 2003; Luo et al., 2024).

S2.1.2 Measurement Uncertainty Assignment

Our compilation included the δ^2H_{wax} value and, where available, the reported uncertainty on this measurement. Where measurement uncertainties were not reported, we assigned an uncertainty of 3‰; where uncertainties were reported as 0‰, we assigned an uncertainty of 1‰. These low uncertainty values are likely to be optimistically precise (Polissar and D'Andrea, 2014), but have little effect on the model since measurement uncertainty on δ^2H_{wax} contributes only minimally to the uncertainty in predicting δ^2H_{precip}.

S2.1.3 Coordinate Reference System

Latitude and longitude data were taken directly from the compilations when reported. Where unstated they were assumed to be reported in a WGS84 (EPSG:4326) coordinate reference system (CRS). Any residual positional uncertainties from undocumented CRS differences are negligible relative to the spatial resolutions used for our calibration. To handle duplicate coordinates that can cause numerical issues in spatial models, we added Gaussian noise with standard deviation 0.0001° (up to 11 meters) to each site's location.

S2.1.4 Environmental Covariate Extraction

We compiled eight global raster datasets as environmental covariates contributing to variations in δ^2H_wax:

1. **δ^2H_{precip}**: Online Isotopes in Precipitation Calculator v3.2 (Bowen and Revenaugh, 2003; IAEA/WMO, 2015; Bowen, 2018) ~10 km resolution
2. **Elevation**: EarthEnv-DEM90 aggregated from CGIAR-CSI SRTM 90m v4.1 (Amatulli et al., 2018)
3. **Plant Functional Type (PFT)**: MODIS MCD12Q1 Land Cover Type 1 product (Friedl and Sulla-Menashe, 2019) averaged 2001-2019, ~9 km resolution
4. **C_4 vegetation fraction**: Luo et al. (2024), global dataset, 1° resolution (~56 km at 30° latitude). For pixels missing from the C_4 contribution raster, we assigned a C_4 fraction of zero to pixels that had |latitudes| ≥ 50° or elevations ≥ 1500, consistent with known biogeography of C_4 plants (Still et al., 2003; Luo et al., 2024).
5. **Climate variables**: TerraClimate (Abatzoglou et al., 2018), ~4 km resolution, 2001-2019 means for:
    - Annual precipitation (sum of monthly values)
    - Soil moisture
    - Maximum temperature
    - Vapor pressure deficit (VPD)

S2.1.5 Multi-scale spatial integration

To account for the spatial integration of leaf waxes in sediment samples, we implemented distance-weighted averaging for environmental predictors (Figure S1). At each sample site *i*, for each spatial buffer scale *r*, we computed an exponentially decaying kernel weight for all grid cells *j* within 1000 km:

$$w_{i,j}^{(r)} = \exp\left(-\frac{d_{i,j}}{r}\right)$$

(Supplementary Equation 5)

where $d_{i,j}$ is the great-circle distance between site *i* and grid cell *j*.

These weights were used to calculate spatially weighted covariate averages:

$$\bar{X}_{i,r} = \frac{\sum_{j \in J_i} \exp\left(-\frac{d_{i,j}}{r}\right) \cdot X_j}{\sum_{j \in J_i} \exp\left(-\frac{d_{i,j}}{r}\right)}$$

(Supplementary Equation 6)

where X_j is the covariate value at grid cell *j* and J_i is the set of all grid cells within 1000 km of site *i.* During model fitting, a single parameter λ controls the effective integration scale by weighting the pre-computed scales:

$$w_r = \frac{\exp(-r/\lambda)}{\sum_{r'} \exp(-r'/\lambda)}$$

(Supplementary Equation 7)

$$\bar{X}_i = \sum_r w_r \cdot \bar{X}_{i,r}$$

(Supplementary Equations 8)

where the sum is over all nine pre-computed scales r' ∈ {1, 3, 5, 10, 20, 40, 70, 100, 150}. This approach effectively interpolates between the pre-computed scales, allowing λ to take any value while leveraging computational efficiency. The λ parameter represents the characteristic distance over which environmental signals are integrated into sedimentary leaf waxes.

S2.1.6 Uncertainty in Environmental Predictors

OIPC provides standard error estimates for δ^2H_{precip} values that were incorporated into our model. Uncertainties in other environmental covariates were not explicitly propagated through the analysis. MODIS land cover classification has an overall accuracy of approximately 75-80% based on classification confidence estimates (Friedl and Sulla-Menashe, 2019). Elevation accuracy in the GMTED2010 dataset is reported as 26-30 meters RMSE for the 7.5 arc-second mean elevation product (Danielson and Gesch, 2011). The C_4 vegetation fraction model reports typical uncertainties of 1-3% of land surface area, rising to 6-7% in certain regions (Luo et al., 2024). TerraClimate has a median absolute error of ~9% against station observations for precipitation (Abatzoglou et al., 2018). However, no spatially explicit error models exist for most of these covariates. Given that their effects on δ^2H_{wax} are comparatively small relative to the dominant spatial patterns, and that quantitative uncertainty models are unavailable for most predictors, these uncertainties were not propagated through the analysis.

S2.2 Exploratory Data Analysis

S2.2.1 Spatial Structure Analysis

We examined the spatial distribution and clustering of samples to understand potential geographic biases in the calibration dataset. The 818 samples showed strong geographic clustering, with high densities in Europe, eastern North America, and East Asia, but sparse coverage in the tropics, central Asia, and oceanic regions (Figure S2).

Nearest-neighbor analysis revealed:

- Mean nearest-neighbor distance: 37 km
- 57.6% of samples within 10 km of another sample
- 92.4% of observations within 100 km of nearest neighbor

To examine whether the δ^2H_{wax}-δ^2H_{precip} relationship varied geographically, we conducted hierarchical clustering on the regression residuals using Ward's method with geographic distance as the similarity metric. This revealed 14 distinct spatial clusters with internally consistent residual patterns, suggesting that regional factors not captured by our environmental predictors systematically influence δ^2H_{wax}.

The strong spatial autocorrelation in OLS residuals (Figure S3; Moran's I = 0.614, $p < 0.001$) persisted across multiple distance classes, with significant positive autocorrelation extending to ~3,000 km. Variogram analysis showed a nugget of 250‰^{2}, partial sill of 400‰^{2}, and range of approximately 3,500 km, consistent with continental-scale spatial processes.

S2.2.2 Multicollinearity Assessment

We computed pairwise Pearson correlations among all candidate predictors (Figure S2). Strong correlations ($|r| > 0.7$) were observed between:

- δ^2H_{precip} and maximum temperature ($r = -0.69$)
- δ^2H_{precip} and VPD ($r = -0.79$)
- Maximum temperature and VPD ($r = 0.86$)
- Annual precipitation and soil moisture ($r = 0.84$)
- Tree fraction and annual precipitation ($r = 0.62$)

Variance inflation factor (VIF) analysis identified several problematic variables:

- VPD: VIF = 12.4
- Maximum temperature: VIF = 10.8
- δ^2H_{precip}: VIF = 8.9 (when all climate variables included)
- Soil moisture: VIF = 6.2

To address collinearity while retaining mechanistic interpretability, we:

1. Removed VPD and maximum temperature (redundant with δ^2H_{precip} signal)
2. Removed soil moisture (highly correlated with annual precipitation)
3. Retained annual precipitation for use in some models
4. Confirmed VIF < 5 for all retained predictors in final model configurations

We also applied elastic net regularization ($\alpha = 0.5$) to assess variable importance under penalization. The regularization path showed that δ^2H_{precip}, elevation, and C_4 fraction were consistently selected even under strong penalties ($\lambda > 0.1$), while climate variables and PFT fractions required weaker penalties to enter the model.

S2.3 Statistical Modeling Framework
S2.3.1 Complete Model Specification
The full hierarchical Bayesian model for sedimentary δ^2H_{wax} is specified as:

$$\delta^2 H_{wax,i} \sim \mathcal{N}(\mu_i, \sigma^2_{total})$$

(Supplementary Equation 9)

where the mean function μ_i includes all fixed and spatially-varying components:

$$\begin{aligned}\mu_i = \beta_0(s_i) + \beta_{OIPC}(s_i) \times \delta^2 H_{precip,i} + f_{elev}(elevation_i) + \beta_{C4} \times C4_i + \beta_{precip} \times precip_i \\ + \beta_{tree} \times tree_i + \beta_{shrub} \times shrub_i + \beta_{grass} \times grass_i \\ + \beta_{OIPC \times C4}\left(\delta^2 H_{precip,i} \times C4_i\right) + \beta_{OIPC \times tree}\left(\delta^2 H_{precip,i} \times tree_i\right) \\ + \beta_{OIPC \times shrub}\left(\delta^2 H_{precip,i} \times shrub_i\right) + \beta_{OIPC \times grass}\left(\delta^2 H_{precip,i} \times grass_i\right)\end{aligned}$$

(Supplementary Equation 10)

S2.3.2 B-spline Specification for Elevation
The elevation effect f_{elev}(elevation) was modeled using cubic B-splines (Wood, 2017):

$$f_{elev}(elevation) = \sum_{j=1}^{K} \beta_{elev,j} \times B_j(elevation)$$

(Supplementary Equation 11)

where:
- K = 13 basis functions (9 interior knots + 4 boundary knots)
- Interior knots placed at quantiles of the elevation distribution
- B_j(elevation) are the B-spline basis functions
- $\beta_{elev,j}$ are the spline coefficients

To address collinearity in the B-spline basis matrix, we centered the basis functions by subtracting the mean of each column, ensuring numerical stability during model fitting.

S2.3.3 Vegetation-Climate Interaction Terms
To test whether vegetation characteristics modulate the relationship between δ^2H_{precip} and δ^2H_{wax}, we implemented interaction terms (Bolker, 2008) between δ^2H_{precip} and vegetation variables. Each $\beta_{OIPC} \times$ veg coefficient represents how much the δ^2H_{precip} slope changes per unit increase in that vegetation type (Jaccard et al., 2003). For example:

- $\beta_{OIPC} \times C_4$ quantifies how the isotopic relationship differs in C_4-dominated ecosystems
- A positive $\beta_{OIPC} \times C_4$ indicates steeper slopes (stronger δ^2H_{wax} response to δ^2H_{precip}) in C_4 vegetation
- A negative value indicates shallower slopes (weaker response) in C_4 vegetation

The interaction terms were pre-computed at each spatial integration scale (1, 3, 5, 10, 20, 30, 50, 70, 100, 150 km) by calculating the spatially weighted products of the variables at each scale. During model fitting, these pre-computed interaction matrices were then weighted by the same exponential kernel process (controlled by parameter λ) as the main effects.

- OIPC×C_4: standardized as the product divided by ($\sigma_{OIPC} \times \sigma_{C4}$)
- OIPC×PFT: standardized as the product divided by σ_{OIPC} (PFTs are proportions 0-1)

S2.3.5 Uncertainty Components

The total variance σ^2_{total} incorporates three sources of uncertainty:

$$\sigma^2_{total} = \sigma^2_{measurement} + \sigma^2_{OIPC} + \sigma^2_{residual}$$

(Supplementary Equation 12)

where:

- $\sigma^2_{measurement}$: Site-specific analytical uncertainty (1-3‰, provided with each measurement)
- σ^2_{OIPC}: Spatial uncertainty in OIPC predictions (5-25‰, location-dependent)
- $\sigma^2_{residual}$: Residual variation capturing micro-scale spatial effects and unexplained variance

S2.3.6 Prior Specifications

We employed weakly informative priors to regularize the model while allowing the data to dominate posterior inference. Prior distributions for all model parameters are detailed in Table S1. The priors were selected based on the following principles:

- Fixed effects: Centered at reasonable values with sufficient width to accommodate uncertainty
- Spatial components: Penalized complexity (PC) priors (Simpson et al., 2017) to control model complexity and prevent overfitting
- Variance components: Weakly informative normal priors to ensure reasonable scale while maintaining flexibility
- Integration scale: Lognormal distribution centered on physically plausible spatial scales for sediment integration

For spatial models, we implemented adaptive regularization where the prior variance for knot effects decreases with local data density, preventing unrealistic parameter values in data-sparse regions (see Section S2.4.5).

S2.3.7 Model Variants

We tested 14 model configurations to evaluate the importance of different components:

- baseline: δ^2H_{precip} only
- baseline_sp: δ^2H_{precip} with spatial effects
- baseline_env: δ^2H_{precip} + elevation
- baseline_env_sp: δ^2H_{precip} + elevation + spatial
- baseline_veg: δ^2H_{precip} + all vegetation
- baseline_veg_sp: δ^2H_{precip} + all vegetation + spatial
- full: All predictors
- full_sp: All predictors + spatial
- full_interact: All predictors + interactions
- full_interact_sp: All predictors + interactions + spatial
- elevation_only_sp: δ^2H_{precip} + elevation + spatial
- elevation_c4_sp: δ^2H_{precip} + elevation + C4 + spatial
- c4_only_sp: δ^2H_{precip} + C4 + spatial
- elevation_c4_interact_sp: δ^2H_{precip} + elevation + C4 + interaction + spatial

S2.4 Spatial Modeling Framework

S2.4.1 Gaussian Process Specification

The spatially-varying intercept $\beta_0(s)$ and slope $\beta_{OIPC}(s)$ follow independent Gaussian processes (GPs; Banerjee et al., 2025):

$$\beta_0(s) \sim \mathcal{GP}\left(\beta_0, k_{\beta_0}(s, s')\right)$$
$$\beta_{\mathrm{OIPC}}(s) \sim \mathcal{GP}\left(\beta_{\mathrm{OIPC,global}}, k_{\beta_{OIPC}}(s, s')\right)$$

(Supplementary Equations 13)

where β_0 and $\beta_{\mathrm{OIPC,global}}$ are global mean values, and k_{β_0} and $k_{\beta_{OIPC}}$ are spatial covariance functions. This formulation allows the intercept and OIPC slope to vary smoothly across space, capturing the environmental heterogeneity observed in our clustering analysis while maintaining the ability to make predictions at new locations.

S2.4.2 Predictive Process Approximation

For computational efficiency, we approximate each GP using the predictive process (Banerjee et al., 2008; Finley et al., 2009):

$$\beta_0(s_i) \approx \boldsymbol{c}_{\boldsymbol{\beta_0}}(s_i)^T \boldsymbol{K}_{\boldsymbol{\beta_0}}^{-1} \boldsymbol{\beta_0}^*$$
$$\beta_{OIPC}(s_i) \approx \boldsymbol{c}_{\boldsymbol{\beta_{OIPC}}}(s_i)^T \boldsymbol{K}_{\boldsymbol{\beta_{OIPC}}}^{-\mathbf{1}} \boldsymbol{\beta_{OIPC}}^*$$

(Supplementary Equations 14)

Where $\boldsymbol{\beta_0}^*$ and $\boldsymbol{\beta_{OIPC}}^*$ are the GP values at $m = 125$ knot locations, $\boldsymbol{K_{\beta_0}}$ and $\boldsymbol{K_{\beta_{OIPC}}}$ are the $m \times m$ knot-to-knot covariance matrices, and $\boldsymbol{c_{\beta_0}}(s_i)$ and $\mathbf{c}_{\boldsymbol{\beta_{OIPC}}}(\mathrm{s_i})$ are $m \times 1$ covariance vectors between site *i* and all knots.

S2.4.3 Matérn Covariance Function
We employed the Matérn 3/2 covariance kernel:

$$k(s_i, s_j) = \eta^2 \left(1 + \frac{\sqrt{3}\, d_{ij}}{\rho}\right) \exp\left(-\frac{\sqrt{3}\, d_{ij}}{\rho}\right)$$

(Supplementary Equation 15)

where:

- η^2 = marginal variance (process amplitude)
- ρ = range parameter (correlation length scale in km)
- d_{ij} = great-circle distance between sites *i* and *j*
-

This kernel balances flexibility and smoothness, allowing the spatial effect to vary smoothly without imposing overly strong assumptions about smoothness (Stein, 1999).

S2.4.4 Knot Placement and Coverage
We optimized the number of knots by balancing two competing considerations: ensuring spatial coverage so that observations can inform the local spatial field, while also recognizing that sparse data inevitably leads to many knots lacking local observations. Our analysis across multiple configurations (75 – 150 knots) indicated that 125 knots achieved a good balance between computational efficiency and spatial resolution. A full formal analysis of knot-number sensitivity was beyond the scope of this study, but the chosen configuration produced stable and interpretable results and was used for all reported models.

We placed knots using a spherical Fibonacci lattice (Swinbank and Purser, 2006) ensuring approximately uniform global distribution. With 125 knots:

- Mean inter-knot distance: ~1,900 km
- 74% of observations within 1,000 km of nearest knot
- 75% of knots had no observations within 1,000 km

This configuration provided adequate spatial resolution for capturing regional climate variations while maintaining computational tractability. The challenge of modeling spatial processes with highly irregular data coverage is well-documented in the analysis of large spatial datasets, where computational methods must balance between capturing local variation in data-rich regions while providing stable predictions in data-sparse areas (Heaton et al., 2019). Our adaptive regularization approach (Section S2.4.5) addresses this challenge by applying stronger constraints to knots in data-sparse regions, preventing overfitting while maintaining the ability to capture genuine spatial patterns where data density permits.

S2.4.5 Spatially-Adaptive Regularization

To prevent overfitting in data-sparse regions, we implemented spatially varying regularization where the prior variance for each knot adapts to local data density. The regularization uses a piecewise function that applies stronger constraints where data are sparse:

For knots with no observations within 1,000 km:

$$\tau_{slope} = 0.40, \quad \tau_{intercept} = 0.50$$

(Supplementary Equation 16)

For knots with 1-10 observations:

$$\tau_{slope} = 0.40 + 0.20 \times \frac{n}{10}, \quad \tau_{intercept} = 0.50 + 0.20 \times \frac{n}{10}$$

(Supplementary Equation 17)

For knots with >10 observations:

$$\tau_{slope} = 0.60 + 0.40 \times \left(1 - e^{-(n-10)/30}\right), \quad \tau_{intercept} = 0.70 + 0.30 \times \left(1 - e^{-(n-10)/30}\right)$$

(Supplementary Equation 18)

This adaptive approach ensures conservative estimates in data-poor regions while allowing the model flexibility to capture genuine spatial patterns where data density permits. The regularization follows the principle that model flexibility should scale with data availability (Simpson et al., 2017; Fuglstad et al., 2019).

S2.4.6 Computational Implementation

Models were implemented in Stan using:

- Non-centered parameterization for hierarchical components
- Cholesky decomposition of covariance matrices
- PC priors on spatial standard deviations (Simpson et al., 2017):
    - Intercept: $P(\sigma > 20‰) = 0.05$
    - Slope: $P(\sigma > 0.3) = 0.05$

Convergence assessed via:

- $\hat{R} < 1.01$ for all parameters
- Effective sample size (ESS bulk) > 900
- No divergent transitions after adaptation*
- E-BFMI > 0.3

* The *baseline_veg_sp* model produced 7 divergences out of 12,000 post-warmup iterations (0.06% divergence rate) distributed across three chains, but was retained as all parameters achieved $\hat{R} = 1.00$ and adequate effective sample sizes, indicating parameter recovery was unaffected. There were no divergences in any other model.

S2.6 Extended Model Validation

S2.6.1 Spatial Cross-Validation Methodology

We implemented leave-one-region-out cross-validation to assess model transferability across continents. For each of five geographic regions (Americas, Europe, Africa, Asia, Oceania), we:

1. Fit the model using all observations except those in the focal region
2. Generated predictions for the held-out region
3. Computed RMSE, mean bias, and coverage statistics
4. Repeated for all regions

This approach is more conservative than leave-one-out cross-validation (LOO-CV) because it tests the model's ability to predict across large geographic distances and different climate regimes. Table S2 provides regional model performance for all model variants.

S2.6.2 Data Density Categories

To evaluate whether model performance degraded in data-sparse regions, we classified each observation by the number of other observations within 0.2 standardized coordinate units (approximately 500 km) of its nearest GP knot:

- Sparse: <10 observations (n=127)
- Medium: 10-50 observations (n=423)
- Dense: >50 observations (n=268)

Performance metrics were calculated separately for each categories. The modest increase in RMSE for dense regions (17.9‰) compared to sparse regions (10.2‰) likely reflects greater environmental heterogeneity in well-sampled areas (e.g., Asian highlands) rather than model failure.

S2.6.3 Overfitting Diagnostics

We assessed three independent indicators of overfitting risk:

Length scale vs. data spacing: Our observations showed strong spatial clustering with 92.4% of sites having a neighbor within 100 km. The estimated GP length scales (3,400-4,250 km) were approximately 2-3 times the mean knot spacing (1,911 km) and 100 times the mean observation spacing, ensuring that the spatial field captures continental-scale trends rather than interpolating between individual sites (Diggle and Ribeiro, 2007).

Effective parameters: The ratio p_{eff}/n provides a measure of model complexity relative to sample size. Values below 0.1 generally indicate appropriate regularization (Piironen and Vehtari, 2017). Our spatial models maintained $p_{eff}/n < 0.07$ across all variants, with the best model at 6.8% (55.9 effective parameters / 818 observations).

Prediction uncertainty: We examined whether prediction intervals appropriately reflected uncertainty by computing the posterior predictive standard deviation at each observation and correlating it with distance to the nearest observation. The spatial model showed appropriately wider intervals in data-sparse regions (range: 57.5‰ to 155.8‰), with mean width of 62.0‰. In contrast, the non-spatial model produced uniformly wide intervals (mean 78.2‰) that failed to capture location-specific uncertainty. No observations had extremely narrow intervals (<5‰), indicating the model avoided overconfident predictions.

S3.3 Spatial Integration

To evaluate whether spatial integration improves model performance independent of the Bayesian framework, we fit the OLS regression (Figure 1) using δ^2H_{precip} values averaged over spatial scales from 0 km (point values) to 100 km with exponential distance weighting. Results (Figure S4) show that spatial integration consistently improves prediction across all modeling approaches, with optimal scales matching Bayesian model estimates.

S3.4 Environmental Predictors

Elevation effects on δ^2H_{wax} showed substantial uncertainty across all models (Figure S5). The B-spline implementation revealed 90% credible intervals spanning ~250. This enormous uncertainty reflects weak constraint from the data once spatial structure is accounted for, indicating that elevation primarily serves as a proxy for regional patterns rather than exerting a strong direct effect on fractionation.

Supplementary Figure captions

Figure S1. Spatial weighting scheme for addressing spatial autocorrelation. (A) Exponential decay kernels showing how observation weights decline with distance for different integration scales (λ). The horizontal dashed line indicates w = 0.37 (1/e), showing the distance at which weights have decayed to ~37% of their maximum value. (B) Example spatial weight fields centered on a site in the central United States for three different integration scales (λ = 20, 70, and 300 km). Warmer colors indicate higher weights.

Figure S2. Pairwise correlations among environmental predictors and response variable. Correlation matrix showing Pearson correlation coefficients between all variables before multicollinearity filtering. Strong correlations ($|r| > 0.7$) are observed between δ^2H_{wax} and several climate variables (OIPC δ^2H_{precip}: r = 0.81; VPD: r = 0.74; max temperature: r = 0.69), and among climate variables themselves (e.g., VPD and max temperature: r = 0.86; annual precipitation and soil moisture: r = 0.84). Variables with high collinearity were excluded from final models to avoid multicollinearity issues (see Methods).

Figure S3. Spatial autocorrelation in OLS regression residuals. Map showing residuals from the ordinary least squares regression of δ²H_wax ~ δ²H_precip (Figure 1). Points are colored by residual magnitude (blue = negative residuals, red = positive residuals). Significant spatial autocorrelation is evident (Moran's I = 0.614, $p < 0.001$; R^2 = 0.65), with clusters of similar residuals indicating that the non-spatial model fails to capture spatially structured variation. This motivates the use of spatial Gaussian process models to account for spatial autocorrelation.

Figure S4. Optimization of spatial integration scale for weighted regression. Model performance (R^2) as a function of spatial integration scale (λ) using exponentially weighted ordinary least squares regression. The red point indicates the optimal scale that maximizes R^2. Performance improves sharply from point values (λ = 0 km, R^2 = 0.653) to the optimal scale (R^2 = 0.701), then plateaus for larger scales. The gray dashed line shows λ = 10 km for reference. This analysis informed the choice of exponential decay weighting in the Gaussian process models.

Figure S5. Elevation effects across models. (A) Estimated effect of elevation on δ^2H_{wax} showing flat effects in non-spatial models and depletion at higher elevations in spatial models. All effects are far smaller than uncertainty. **(B)**. Histogram of the number of samples at various elevations. Bin width = 250 m.

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